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Professional Overview:

- Over **12 years** of experience as a **Senior Python Developer**, specialized in AI/ML, web services and data processing with technologies such as **TensorFlow**, **Django** and **FastAPI**. Expertise in the SDLC, RESTful API development and big data solutions using **PySpark**. Skilled in database management, cloud computing across **AWS**, **GCP**, **Azure** and CI/CD implementations with **Git** and **Jenkins**. Leadership in development teams, advocating for best practices and driving innovation in Python-based projects.
- Demonstrated mastery in Python, utilizing advanced libraries such as **TensorFlow**, **PyTorch** for AI/ML applications, **Django** and **Flask** for web development and **asyncio** for efficient asynchronous programming. My ability to design custom Python modules addresses specialized needs across projects.
- Experienced in the entire Software Development Life Cycle (SDLC) with Python, adept at navigating through system requirements analysis, architecture design, development, testing and maintenance. My versatility shines in both **Agile** and **Waterfall** methodologies, adapting processes to best fit Python-centric projects.
- Proficient in an array of Python frameworks; I leverage **Django** for comprehensive web development projects, **Flask** and **FastAPI** for crafting performant APIs and **Pyramid** for its flexibility in various applications. My expertise extends to using **Tornado** for asynchronous apps and **Bottle** and **CherryPy** for quick, minimalistic solutions, showcasing my adaptive Python development capabilities.
- Skilled in employing Python for impactful data visualization and reporting, making extensive use of libraries like **Matplotlib**, **Seaborn** and **Bokeh**. This proficiency enables the delivery of meaningful data insights through visually appealing presentations and interactive dashboards.
- My capabilities in optimizing Python applications for performance include automating and refining ETL processes and integrating **PySpark** and **Apache Beam** for big data processing, thereby creating scalable data pipelines and performing advanced analytics.
- Specialize in designing and implementing RESTful APIs with **Flask** and **FastAPI**, ensuring scalability, efficiency and security. Employ **OAuth2** and **JWT** for authentication and use **Swagger/Open API** for clear documentation, facilitating seamless data integration and optimal application performance.
- With a solid background in machine learning and data analysis, I harness **Pandas**, **NumPy**, **SciPy** and **Scikit-learn**, complemented by **TensorFlow** and **Keras**, for the development and deployment of machine learning models, offering sophisticated data-driven solutions.
- My experience spans working with both SQL and NoSQL databases via Python, utilizing ORM tools like **SQLAlchemy** and engaging with **MongoDB**, **Cassandra** and **DynamoDB** to ensure dynamic data storage and retrieval options.
- Adept in cloud computing environments, I deploy Python applications across **AWS**, **GCP** and **Azure**, maximizing the potential of cloud-specific SDKs and APIs for streamlined integration and robust automation solutions.
- My hands-on approach with CI/CD pipelines in Python projects employs **Git** for version control and **Jenkins**, **Terraform** and **Ansible** for sophisticated automation and deployment practices, ensuring a seamless delivery process.
- Proficient in advanced **Git** version control, skilled in managing complex codebases, adept in branching strategies like **Git Flow** and **Trunk Based Development** and experienced in code reviews and pull request management using platforms like **GitHub**, **GitLab** and **Bitbucket**.
- Committed to continuous improvement and knowledge sharing, I engage in mentoring and leading teams, advocating for best practices in Python development and staying updated with the latest technological advancements to drive strategic innovations and solutions.

Technical Skills:

Category	Tools/Technologies
Programming Language	Python, Java
AI/ML Libraries	TensorFlow, PyTorch
Web Development Frameworks	Django, Flask, FastAPI
Asynchronous Programming	asyncio
Big Data Processing	PySpark, Apache Beam, Hadoop
Data Visualization Libraries	Matplotlib, Seaborn, Bokeh
Database Management	SQLAlchemy, MongoDB, Cassandra, DynamoDB
Cloud Computing Platforms	AWS, GCP, Azure
CI/CD Tools	Git, Jenkins, Terraform, Ansible
API Development & Documentation	Flask, FastAPI, Swagger/Open API
Authentication & Security	OAuth2, JWT
Data Analysis & Machine Learning	Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras
Version Control & Collaboration	Git, GitHub, GitLab, Bitbucket
SDLC Methodologies	Agile, Waterfall
Miscellaneous Frameworks	Pyramid, Tornado, Bottle, CherryPy

Scholarly Pursuits:

Certified Achievements:

- **AWS** Certified Developer Associate.
- Microsoft Certified **Azure** Developer Associate.

Work Portfolio:

Client: CVS Health, Chicago – IL **Sept 2022 – Present**
Senior Python Developer
Functional Role Details:

- Develop and maintain large-scale data processing applications in healthcare using Python, leveraging Pandas, NumPy, and Matplotlib for advanced analytics. Designed and built RESTful APIs and JSON APIs to facilitate seamless data exchange between distributed microservices and external healthcare systems. Developed dynamic, interactive user interfaces using Vue.js for real-time health data management and analytics dashboards.
- Architect and optimize backend processing pipelines by integrating legacy healthcare systems with modernized data platforms, utilizing PHP 7-5 and optimized MySQL queries for high-throughput data ingestion and processing. Leveraged Linux shell scripting to automate complex data validation, ingestion, and compliance workflows, ensuring regulatory standards adherence.
- Develop scalable ETL/ELT workflows using Python, PySpark, and Informatica, managing high-volume healthcare datasets with distributed data transformation processes. Designed high-performance data models across relational (Oracle, MySQL, PostgreSQL) and NoSQL (Couchbase) databases, ensuring optimized data storage, retrieval, and compliance adherence. Built and orchestrated Snowflake-based pipelines, leveraging cloud-native capabilities for scalable analytics and real-time decision-making.
- Lead the development of cloud-native data solutions on AWS, orchestrating ETL processes with Python scripts, Informatica, and PySpark for real-time analytics. Implemented AWS services like S3, Lambda, and RDS for efficient data

processing and scalable infrastructure deployment. Integrated Snowflake-based analytics pipelines to optimize enterprise-wide insights and predictive healthcare models.

- Design and implement large-scale distributed computing solutions using Hadoop and PySpark, managing big data pipelines and optimizing performance with Hive for structured and semi-structured data querying. Designed highly optimized REST API endpoints to handle structured and unstructured data processing, ensuring high availability and fault tolerance in mission-critical healthcare applications.
- Enhance data governance and real-time monitoring using Apache NiFi, Informatica, and Hadoop-based architectures. Developed interactive Vue.js applications integrated with Apache Airflow, providing real-time monitoring and control over batch processing workflows. Automated ETL execution scripts using Linux shell scripting, reducing manual intervention and optimizing resource utilization.
- Architect and deploy financial-grade healthcare analytics solutions by integrating Python-based data pipelines with Snowflake, ensuring real-time insights across complex datasets. Implemented SQL-based data transformation logic for high-performance analytics queries. Developed RESTful API integrations to enable real-time data exchange with external partners and internal analytics platforms.
- Lead integration of enterprise-wide data processing platforms using Python, Informatica, and Hadoop ecosystems, ensuring scalable and secure data exchange. Developed highly optimized REST APIs for secure, low-latency financial transactions and healthcare claims processing. Built Vue.js-based UI applications for interactive API monitoring and real-time data visualization.
- Develop high-performance machine learning pipelines using Python, PySpark, Keras, and TensorFlow, enabling predictive analytics and AI-driven healthcare insights. Designed Vue.js-based MLOps interfaces for real-time model monitoring, deployment, and performance tracking. Integrated AWS ML services for cloud-scale training and inference workflows, ensuring scalability and high availability.
- Spearhead DevOps and automation initiatives using Python scripting, Git, and CI/CD pipelines, ensuring high code quality and continuous integration workflows. Developed Java-based API modules for high-performance data ingestion, optimizing real-time data processing workflows. Automated log analysis and incident management using Humio, Logscale, and Python-based alerting systems.
- Strengthen application security and compliance by implementing Python-based security frameworks with Vault for secrets management, ensuring secure API authentication and regulatory compliance. Deployed Linux-based security hardening strategies and established incident response workflows for intrusion detection and data protection.
- Architect containerized microservices solutions using Docker and Kubernetes, enabling high-availability, scalable deployment of healthcare applications. Developed Golang-based container orchestration scripts to enhance system reliability and optimize resource allocation. Managed Linux server environments for high-performance data-intensive applications.
- Develop automated reporting and financial-grade analytics dashboards using Python, Informatica, and reporting frameworks like Jasper Reports. Built interactive Vue.js-based front-end interfaces for custom report generation and real-time data exploration. Designed optimized Snowflake-based data models to support financial, regulatory, and operational reporting requirements.

Project Infrastructure: Pandas, NumPy, Matplotlib, Django, Flask, ER/Studio, PowerDesigner, Liquibase, Flyway, PySpark, Kafka, AWS Glue, Amazon RDS (PostgreSQL/MySQL/Aurora), Amazon S3, Apache NiFi, Informatica, Apache Airflow, MongoDB, dbt, Amazon Redshift, Talend, Tableau, D3.js, Looker, Collibra, Data Cleaner, Prometheus, Keras, MLflow, Git, PowerShell, Jasper Reports, Vault, Docker, Kubernetes, JUnit.

Client: SMBC, New York, NY **Nov 2020 – Aug 202**

Senior Python Developer

Functional Role Details:

- Masterfully developed Python scripts, seamlessly integrating with Azure Databricks for efficient banking data processing within Data Lake and Blob Storage. Utilized PySpark data frames and Spark SQL for transformative data analysis and reporting.

- Engineered sophisticated data integration strategies for banking analytics using Python, leveraging Azure Data Lake Analytics, Data Lake Storage, Data Factory, Azure SQL Databases and SQL Data Warehouse for comprehensive data management.
- Orchestrated complex data migration projects within the banking sector using Python, facilitating secure database transitions to Azure Data Lake Store and integrating banking systems with MongoDB and MS SQL via Azure Data Factory.
- Used AWS SageMaker to create a real-time fraud detection system for banking transactions, incorporating machine learning models for preventative fraud and regulatory adherence.
- Automated real-time transaction monitoring system for banks uses Hadoop, Flink, Apache Kafka, and HBase, with Ansible playbooks for infrastructure provisioning, reducing human labor, and accelerating fraud detection.
- Implemented change data capture (CDC) methodologies in data pipelines using Python, enhancing data transformation processes with Azure Data Factory. Analyzed transactional data stored in HDFS using Hive, Presto and Spark for insightful reporting.
- Loaded transactional data into partitioned Hive tables from banking systems, including Teradata, using Python and Sqoop. Optimized HiveQL for fast data retrieval, aiding in analysis and decision-making.
- Designed data integration frameworks for the banking industry using Python, ensuring seamless data exchange between disparate banking systems and platforms for improved operational efficiency.
- Utilized Azure Synapse Analytics with Python scripting to optimize data processing workloads, supporting advanced business intelligence (BI) and predictive analytics in banking.
- Established stringent data governance frameworks in banking applications using Python, Azure Data Lake Storage, Azure Active Directory and Azure Key Vault, ensuring compliance with regulations and data security.
- Developed an Enterprise Data Lake for banking analytics using Azure Data Factory and Blob Storage, integrating Python to enable sophisticated Machine Learning (ML) algorithms for fraud detection and customer insights.
- Facilitated secure and efficient data exchanges between on-premises banking systems and cloud environments using Python with Azure Data Factory's Copy Activity, employing Azure Synapse Analytics for optimal data management.
- Leveraged ReactJS for the development of SPA banking portals, focusing on user engagement and real-time data presentation. Implemented state management solutions using Redux and Context API to manage complex application states across the banking platform.
- Used React hooks for functional components to build responsive and dynamic web interfaces, ensuring optimal performance and user interaction in banking applications.
- Enhanced the banking application's front-end architecture with ReactJS, focusing on modular design and component reuse to streamline development and maintenance processes.
- Configured Snowflake within the Azure ecosystem for banking data warehouses, using Python to improve data storage and analysis capabilities.
- Crafted and maintained ETL/ELT workflows in banking systems with Python, employing SSIS and Informatica to maintain high data quality and consistency.
- Developed SSIS packages focusing on complex banking data transformations with Python, streamlining ETL operations for data management.
- Designed and deployed interactive banking interfaces using Angular, enhanced with HTML, CSS and JavaScript, to provide intuitive data visualization and customer engagement in Python-based banking applications.
- Managed Python codebases for banking projects using Git for version control, facilitating collaborative development and efficient code management in data engineering initiatives.
- Toolkit & Platform: Python, Azure Databricks, PySpark, Spark SQL, Azure Data Lake Analytics, Azure Data Lake Storage, Azure Data Factory, Azure SQL databases, SQL Data Warehouse, MongoDB, MS SQL, HDFS,

Hive, Presto, Teradata, Sqoop, HTML, CSS, JavaScript, Angular 10+, Azure Synapse Analytics, Azure Active Directory, Azure Key Vault, Blob Storage, Snowflake, SSIS, Informatica and Git.

Client: Discover Financial, River woods, IL **Jan 2019 to Oct 2022**
Senior Python Developer
Functional Role Details:

- Conceptualized and implemented robust ETL pipelines using Python, Spark and PySpark, enhancing data processing automation for financial analytics and reporting.
- Integrated a variety of relational databases such as MySQL, Teradata, Oracle, Sybase, SQL Server and DB2 using Python, laying the groundwork for comprehensive data warehousing and integration strategies within the financial sector.
- Employed Airflow to orchestrate complex ETL sequences, significantly improving data workflow management in financial data processing.
- Spearheaded the migration from on-premises Hadoop ecosystems to cloud-based solutions in GCP, utilizing Python to ensure seamless data workflows and integration for financial services.
- Utilized Grafana for visual metric representations, extracting data from Cassandra databases and led the migration of Oracle databases to BigQuery in GCP for advanced financial analytics.
- Advanced data visualization techniques were deployed using Python with data in BigQuery, facilitating interactive data storytelling and decision-making in finance.
- Established stringent data quality frameworks using Python, focusing on accuracy, uniformity and completeness, crucial for financial data integrity.
- Initiated comprehensive data stewardship protocols with Python, defining clear data ownership, access rules and lifecycle management in financial systems.
- Collaborated closely with data analysts and machine learning specialists, utilizing Python to support sophisticated analytics and predictive modeling in finance.
- Expertly managed source code using Git, enhancing collaborative development and code quality in financial software projects.
- Transitioned towards Full-Stack development with a focus on front-end technologies, particularly ReactJS, to revamp Discover Financial's web applications. Developed responsive and interactive user interfaces using ReactJS, creating seamless integration with Python backend services for dynamic data interactions and real-time financial analytics.
- Employed ReactJS to build single-page applications (SPAs) for financial services, enhancing user experience with modern web technologies. Utilized state management techniques with Redux and context API in React to manage complex application states, improving the performance and usability of financial web applications
- Enhanced the financial application's front-end architecture with ReactJS, focusing on component-based design to streamline development and maintenance. This initiative not only accelerated the development cycle but also ensured high code quality and reusability across Discover Financial's web projects.
- Crafted responsive and dynamic user interfaces for financial applications using the Angular framework, HTML, CSS and JavaScript, integrated with Python backend services for seamless data interaction.
- Led the optimization of Linux and Windows operating systems for data engineering tasks in financial environments, ensuring system efficiency and reliability.
- Expertly integrated React hooks and functional components to develop highly responsive and dynamic financial interfaces. This approach facilitated efficient data presentation and user interactions, catering to the specific needs of financial clients and stakeholders.
- Authored and maintained sophisticated automation scripts with PowerShell and Python, streamlining financial data operations and system administration.

- Toolkit & Platform: Python, MySQL, Teradata, Oracle, Sybase, SQL Server, DB2, Spark, PySpark, Airflow, GCP, Grafana, Cassandra, BigQuery, Teradata SQL, Tableau, Git, Linux, Windows and PowerShell.

Client: High Radius Technologies, India **Apr 2016 –Nov 2018**
Python Developer
Functional Role Details:

- Demonstrated advanced proficiency in Python, leveraging libraries like Pandas for efficient data manipulation, analysis and automation, improving overall data processing and performance in backend services.
- Developed scalable APIs using Django and Flask, enhancing system interoperability, scalability and seamless communication between distributed systems and microservices.
- Integrated third-party APIs and services into Python applications, utilizing Requests and HTTPX, ensuring reliable data retrieval and seamless integration with external platforms.
- Leveraged Google Cloud Services to deploy and manage Python applications, utilizing Google Cloud Functions and Google Cloud Pub/Sub for real-time, event-driven architectures.
- Automated cloud infrastructure management using Terraform and Google Cloud Deployment Manager, utilizing Python for Infrastructure as Code (IaC) to ensure scalable and consistent provisioning of resources.
- Deployed serverless Python applications with Google Cloud Functions, enhancing system scalability and reducing operational overhead by integrating real-time event triggers and background processing.
- Managed cloud services using Python SDKs, interacting with core services like Google Compute Engine (GCE), Google Cloud Storage and Google Cloud Filestore to ensure optimized resource management and secure operations.
- Developed machine learning models using Python libraries like Scikit-Learn and TensorFlow, improving predictive analytics and driving AI-driven innovations.
- Implemented CI/CD pipelines using Jenkins and Python scripting, automating development, testing and deployment processes for faster, more reliable release cycles.
- Optimized backend performance using Redis caching, reducing database load and delivering faster response times for high-traffic applications.
- Implemented secure authentication and authorization mechanisms in Python applications using OAuth2 and JWT tokens, ensuring secure access to APIs and protecting sensitive user data.
- Orchestrated workflows with Prefect, automating task scheduling and execution in Python applications to improve system efficiency and reliability.
- Collaborated with front-end teams to integrate Python backends with web applications built using Angular, ensuring seamless data exchange and enhanced user experiences.